

CAREER OVERVIEW

I am a Lecturer in AI and Data Science with a research and teaching profile. My research uses computer vision, embedded AI, and human–computer interaction to support independent living and ageing in place. My work spans human activity recognition and user acceptance of ambient assisted living technologies. I have published in high-quality international journals and conferences, including venues with high impact factors, and my work has been recognised through best paper awards. Alongside my research, I have experience in teaching and leading modules in AI and Data Science, with a commitment to inclusive, research-informed teaching and student success.

RESEARCH INTEREST

My research interests are multi-disciplinary and cut across the following fields:

- Computer Vision (Artificial Intelligence)
- Human Computer Interaction (HCI)
- Ambient Assisted Living Technologies (AAL)
- Smart and Sustainable Home Environment

EDUCATION

- **Doctor of Philosophy (PhD)** in Applied Artificial Intelligence, Department of Applied Artificial Intelligence and Robotics, Aston University, UK. Oct. 2022 – Sept. 2025
 - *Published 11 high-impact, peer-reviewed articles during my PhD.*
- **Postgraduate Certificate Learning and Teaching in Higher Education (PGCert)**, Aston University, UK. Awarded Fellowship of the Higher Education Academy (**FHEA**) by **AdvanceHE**. Oct. 2023 – Nov. 2024
- **Introduction to Learning and Teaching Practice in Higher Education (ILTP)**, Aston University, UK. Awarded Associate Fellowship of the Higher Education Academy (**AFHEA**) by **AdvanceHE**. March 2023 – Sept. 2023
- **Master of Engineering (Research) – Mechatronics Engineering**, Department of Mechatronics Engineering, Zhejiang University, China. (*QS Top 49 ranked University, 2026*) 2018 – 2021
 - *Published two high-impact, peer-reviewed articles based on my Master's thesis; one article was **featured on the journal's front cover**.*
- **Bachelor of Engineering – Electronics Information Engineering (First Class Honors)**, Department of Electronic and Information Engineering, Liaoning University of Technology, China. 2014 – 2018
 - *I was the best student for 4 consecutive years in my department.*
 - *I was the best graduating student with **GPA of 4.8** out of 5.0.*

PROFESSIONAL EXPERIENCE

Lecturer in AI and Data Science: University of Hull

June. 2025 – Present

- Actively involved in research and teaching.
- Module leader (Level 7) for Programming for Artificial Intelligence and Data Science and Understanding Artificial Intelligence. Average class size of 45 students.

- Module Lecturer for Programming for Artificial Intelligence and Data Science, Understanding Artificial Intelligence, Fundamental of Data Science, and Applied Artificial Intelligence (All Level 7). Average class size of 45 students.
- Involved in teaching material development, delivery and providing feedback on assessments.
- Postgraduate Research/Dissertation supervision.
- Provide pastoral support for students as personal tutor.
- Involved in research and publishing in high-impact journals, and conferences.

External Academic Collaborator (AI Research & Doctoral Supervision): The Hong Kong Polytechnic University.

Dec. 2025 – Present

- External collaboration on AI-driven research in occupant-centric building control.
- Co-supervision of PhD student, providing expertise on application of AI in Solar Photovoltaic post disaster inspection.
- Contribution to research development and academic publications.

Lecturer in Computing: Ulster University (QA Higher Education)

Jan. 2025. – Nov. 2025

- Module tutor for Scalable Advanced Software Solutions (Level 7). Average class size of 50 students.
- Involved in teaching material development, delivery and providing feedback on assessments.
- Postgraduate Research/Dissertation supervision.

Postgraduate Teaching Assistant: Aston University

Oct. 2022 – Sept. 2025

- I assisted senior academics in teaching and supporting active learning for both undergraduate and postgraduate students.
- Assessed coursework, exams, and projects, providing detailed, constructive feedback to enhance student learning
- Postgraduate Research/Dissertation supervision.
- Delivered tutorials and practical sessions for the following modules: Software Engineering (Level 5), Software Project Management (Level 6), Introduction to Artificial Intelligence (Level 5), Data Mining/Understanding Data (Level 7), and Machine Learning (Level 7).

Research Associate: The Hong Kong Polytechnic University, Hong Kong

Nov. 2021 – July 2022

I Worked on the funded project "*Human-Centred Smart and Sustainable Building Management System.*" The roles included:

- **Research:** Undertook primary responsibility for the development of my research and contributed to international and local research collaborations. Provided technical support for research students within the research group. Involved in grant applications.
- **Hardware Procurement:** Managed procurement of devices, evaluating options and ensuring cost-effective, timely delivery.
- **Research Outputs:** Developed solutions for human-centred building control using computer vision. Wrote and published research papers in top-rated journals.
- **Administration:** I worked alongside the Principal Investigator for the overall management of the research group.
- **Research group website:** I designed and managed the research group's website (<https://ibeems-lee.com/>)

Research Associate: Westlake University, China.

June 2021 – Nov. 2021

I worked on the funded project "*Ecosystem Monitoring Project*" at Westlake University. My roles included:

- **Hardware Development:** I designed and developed electronic hardware components for a computer vision-based camera system to monitor agrobiodiversity.
- **Procurement Management:** I managed the procurement of devices, coordinating with vendors to ensure cost-effective and timely delivery.
- **Field Testing:** I collaborated with Ecologists to integrate and validate the system in real-world environments.
- **Team Collaboration:** I worked with multidisciplinary teams, including software designers, Ecologists and AutoCAD specialists, to deliver a comprehensive solution.
- **Research Contributions:** Contributed to research publication on Embedded vision cameras for terrestrial biodiversity monitoring.

FUNDED RESEARCH PROJECTS

Human-Centered Smart and Sustainable Building Management System at the Room, Building, and Community Level.
Funding: HK\$ 600,000, Department of Building and Real Estate, The Hong Kong Polytechnic University, Hong Kong.
Duration: Dec. 2020 – Aug. 2024. **Status:** Completed. **Role:** Co-Investigator.

Embedded vision cameras for terrestrial biodiversity monitoring. **Funding:** RMB 800,000, School of Engineering, Westlake University, China. **Duration:** Jan. 2020 – Aug. 2023. **Status:** Completed. **Role:** I joined the team and worked on the design and development of electronic hardware components for the computer vision-based camera system.

Human Activity Analysis in Smart Environments. **Funding:** Fully funded PhD Studentship awarded by UK Research and Innovation (UKRI). **Institution:** Aston University. **Duration:** Oct. 2022 – Sept. 2025. **Status:** Completed. **Role:** Doctoral Researcher.

PROFESSIONAL CERTIFICATION

- **AdvanceHE:** Fellowship of the Higher Education Academy (FHEA) Nov. 2024
- **AdvanceHE:** Associate Fellow of the Higher Education Academy (AFHEA) Sept. 2023

AWARDS

- **Best Paper Runner-Up:** International Conference on AI in Healthcare 2024
- **Best Paper Award:** 22nd International Conference on Construction Applications of Virtual Reality 2022
- **UK Research and Innovation (UKRI) PhD Studentship:** Fully funded doctoral research on Human activity analysis in smart environments, awarded by UKRI. 2022–2025
- **F.C.T Scholarship Board:** Ministerial Special Scholarship Award: Award of Excellence 2018
- **Liaoning University of Technology:** Best Student in Electronic and Information Engineering Department 2016/2017 Academic Year
- **Liaoning University of Technology:** Second Best Student in Electronic and Information Engineering Department 2015/2016 Academic Year
- **Liaoning University of Technology:** Second Best Student in Electronic and Information Engineering Department 2014/2015 Academic Year
- **F.C.T Scholarship Board:** Best Indigene Student in West African Senior School Certificate Examination 2013

STUDENT SUPERVISION

Doctoral Supervision (Ongoing)

- **Student Name:** Wenjun GAO
Role: Co-supervisor (External)
Institution: The Hong Kong Polytechnic University
Topic: Application of Artificial Intelligence in Solar Photovoltaic (PV)

Master's Dissertation Supervision (Selected)

The following is a representative selection of Master's dissertation students supervised.

- **Student Name:** Maimunatu Tunau
Project: Sparse Point Cloud Data Processing for Privacy-aware Human Action Recognition
Institution: Aston University
Outcome: Distinction; resulted in peer-reviewed publication (https://link.springer.com/chapter/10.1007/978-3-032-00656-1_11)
- **Student Name:** Mehedi Hassan
Project: Improving Trust in AI-Based Histopathology: Transfer Learning and Explainability for Breast Cancer Image Classification
Institution: Ulster University
Outcome: Distinction
- **Student Name:** Min Thurein Htet
Project: Multi-Label PPE Classification in Construction Images Using MobileNetV2 Transfer Learning
Institution: University of Hull
Outcome: Distinction
- **Student Name:** Amos Okpe Orevaoghene
Project: Long-Term Multi-Person Action Recognition and Tracking
Institution: University of Hull
Outcome: Distinction

PROFESSIONAL SERVICE

Peer Reviewing

- IEEE/RSJ International Conference on Intelligent Robots and Systems
- Proceedings of Machine Learning Research
- International Conference on AI in Healthcare
- Journal of Technovation

Organising Committee

The Fourth UK AI Conference 2026 ([Index | The Fourth UK AI Conference 2026](#))

Editorial Role

Editor, UK AI Proceedings in Proceedings of Machine Learning Research (<https://proceedings.mlr.press/v295/>)

PUBLICATIONS (IF = Impact Factor)

1. Action Recognition in Real-World Ambient Assisted Living Environment. **Vincent Gbouna Zakka**, Zhuangzhuang Dai, Luis J. Manso. Journal of Big Data Mining and Analytics: <https://doi.org/10.26599/BDMA.2025.9020003>. (IF: 6.2)
2. GazeMoE: Perception of Gaze Target with Mixture-of-Experts. Zhuangzhuang Dai, Zhongxi Lu, Vincent G Zakka, Luis J Manso, Jose M Alcaraz Calero, Chen Li. IEEE International Conference on Robotics & Automation: <https://arxiv.org/abs/2603.06256>. (IF = N/A, Conference publication)

3. GazeTarget360: Towards Gaze Target Estimation in 360-Degree for Robot Perception. Dai, Zhuangzhuang and **Zakka, Gbouna Zakka**, and Manso, Luis J. and Li, Chen. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): <https://arxiv.org/abs/2507.00253>. (IF = N/A, Conference publication)
4. Enhanced Sparse Point Cloud Data Processing for Privacy-aware Human Action Recognition. Maimunatu Tunau, **Vincent Gbouna Zakka**, Zhuangzhuang Dai. International Conference on AI in Healthcare: https://link.springer.com/chapter/10.1007/978-3-032-00656-1_11. (IF = N/A, Conference publication)
5. Analysing different household energy use patterns using clustering and machine learning. Xue Cui, Minhyun Lee, Mohammad Nyme Uddin, Xuange Zhang, **Vincent Gbouna Zakka**. Journal of Renewable and Sustainable Energy Reviews. <https://doi.org/10.1016/j.rser.2025.115335>. (IF: 16.3)
6. Hierarchical Temporal Convolution Network: Towards Privacy-Centric Activity Recognition. **Vincent Gbouna Zakka**, Zhuangzhuang Dai, Luis J. Manso. 16th International Conference on Ubiquitous Computing and Ambient Intelligence: https://doi.org/10.1007/978-3-031-77571-0_33. (IF = N/A, Conference publication)
7. Sensors, Techniques and Future Trends of Human Engagement Enabled Applications: A Review. Zhuangzhuang Dai, **Vincent Gbouna Zakka**, Luis J. Manso, Martin Rudorfer, Ulysses Bernardet, Johanna Zumer, Manolya Kavakli-Thorne. Journal of Algorithms: <https://doi.org/10.3390/a17120560>. (IF: 2.1)
8. Action Recognition for Privacy-Preserving Ambient Assisted Living. **Vincent Gbouna Zakka**, Zhuangzhuang Dai, Luis J. Manso. International Conference on AI in Healthcare: https://doi.org/10.1007/978-3-031-67285-9_15. (IF = N/A, Conference publication). **Best Paper Award.**
9. A computer vision-based model for occupancy detection using low-resolution thermal images. Xue Cui, Minhyun Lee, **Vincent Gbouna Zakka**. 6th International Conference on Building Energy and Environment. Accepted paper on arXiv: <https://arxiv.org/abs/2505.08336>. (IF = N/A, Conference publication)
10. Eyes on nature: Embedded vision cameras for terrestrial biodiversity monitoring. Darras, Kevin; Balle, Marcel; Xu, Wenxiu; Yan, Yang; **Zakka Gbouna, Vincent**; Toletdo, Manuel; Sheng, Dong; Lin, Wei; Zhang, Boyu; Lan, Zhenzhong; Fupeng, Li; Wanger, Thomas. Method in Ecology: <https://doi.org/10.1111/2041-210X.14436>. (IF: 6.2) **(Top viewed article)**
11. Non-invasive vision-based personal comfort model using thermographic images and deep learning. **Vincent Gbouna Zakka**, Minhyun Lee, Ruxiaoxiao Zhang, Lijie Huang, Seunghoon Jung, Taehoon Hong. Automation in Construction: <https://doi.org/10.1016/j.autcon.2024.105811>. (IF: 11.5)
12. A generalized thermal comfort model using thermographic images and compact convolutional transformers Towards scalable and adaptive occupant comfort optimization. **Vincent Gbouna Zakka**, Minhyun Lee. Building and Environment: <https://doi.org/10.1016/j.buildenv.2024.112118>. (IF: 7.6)
13. The impact of personal environmental control on the performance of thermal systems: Building energy consumption, occupant thermal comfort and productivity. Kexin Xie, Minhyun Lee, Rihab Khalid, **Vincent Gbouna Zakka**. Energy and Buildings: <https://doi.org/10.1016/j.enbuild.2023.113552>. (IF: 7.1)
14. An Integrated Design of Energy and Indoor Environmental Quality System for Effective Building Performance Management. **Vincent Gbouna Zakka**, Minhyun Lee. Proceedings of the 22nd International Conference on Construction Applications of Virtual Reality (CONVR 2022) (pp. 455-466): <https://arxiv.org/abs/2503.23323>. (IF = N/A, Conference publication). **Best Paper Award.**
15. User-interactive robot skin with large-area scalability for safer and natural human-robot collaboration in future telehealthcare. **Zakka Vincent Gbouna**, Gaoyang Pang, Geng Yang, Zeyang Hou, Honghao Lv, Zhangwei Yu, Zhibo Pang. IEEE Journal of Biomedical and Health Informatics: <https://doi.org/10.1109/JBHI.2021.3082563>. (IF: 6.7) **(Featured on Cover Paper)**
16. IoT-Enabled Robot Skin System for Enhancement of Safe Human-Robot Collaboration. **Zakka Vincent Gbouna**, Zeyang Hou, Gaoyang Pang, Huayong Yang, Geng Yang. International Conference on Intelligent Robotics and Applications: https://doi.org/10.1007/978-3-030-89098-8_43. (IF = N/A, Conference publication)
17. Facile Fabrication of Highly Soft Tactile Sensor based on Porous Sponge with Geometry Effect on Sensing Characteristics. Wenzhou Heng, Gaoyang Pang, **Zakka Vincent Gbouna**, Zhibo Pang, Kang Yi, Huayong Yang, Geng Yang. 2nd IEEE International Flexible Electronics Technology Conference: <https://doi.org/10.1109/IFETC46817.2019.9073774>. (IF = N/A, Conference publication)